

Title

Comparable efficacy of over-the-counter salicylic acid system and prescription retinoid combination for mild-moderate acne

Abstract

Objective: To compare efficacy and tolerability of a cosmetic acne regimen with prescription therapy in diverse populations. **Design:** Single-center, double-blind trial. **Participants:** 54 adults (31 cosmetic regimen, 23 prescription regimen), predominantly female (90.7%), diverse ethnic backgrounds. **Intervention:** Cosmetic salicylic acid system versus prescription retinoid-peroxide combination applied to face for 12 weeks. **Results:** Clinician-assessed acne severity improved significantly from baseline in both groups beginning at 48 hours and sustained through 12 weeks ($p \leq 0.001$). Total facial lesion counts declined similarly in both treatment arms. Post-inflammatory changes and facial oiliness showed comparable improvement. Notably, the cosmetic regimen demonstrated significantly better skin tolerability scores at week 4 for skin tightness and dryness sensations ($p = 0.008$). Patient satisfaction exceeded 96% with cosmetic regimen across all timepoints versus lower satisfaction with prescription therapy due to irritation concerns. Both regimens were well-tolerated overall, with no serious adverse events. **Conclusions:** The cosmetic anti-acne regimen achieved comparable efficacy to prescription therapy with superior tolerability in patients with mild-to-moderate acne.